

Creating a Podcast Using Generative Artificial Intelligence to Support Student Learning of Organic Reaction Mechanisms

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ETHICS

The podcast was developed following standard university policy, which strongly encourages instructor-led use of Google AI tools in innovative classroom teaching interventions without the need for formal ethical approval. Student feedback was collected through a carefully designed survey tool. Informed consent for data collection in the survey tool was obtained in accordance with institutional policies, which for feedback on classroom activities are devolved to the Departmental Education Committee, who provided oversight. The survey was delivered on an opt-in basis, with all participants being made aware prior to participation that survey data and qualitative feedback would potentially be used as part of a publication.

TRANSCRIPTS OF ORIGINAL PODCASTS

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PODCAST VERSION 1

USER PROMPT:

Explain why having rules to predict organic reaction mechanisms is important and useful then describe all of the rules for predicting organic reaction mechanisms and discuss how to apply them.

AI OUTPUT

Cracking the Code: The 5 Algorithmic Rules to Predict Organic Chemistry Mechanisms (No Memorization Required)

Okay. So, if you've ever looked at organic chemistry

or maybe uh you're steering it down right now, you probably know that feeling, that crushing weight of just memorization.

Oh, yeah.

Mechanisms pile up, dozens of them, and those curly arrows, that electron pushing stuff often feels like, you know, mystical scribbles. Yeah. You just have to copy blindly to get the answer.

That's such a common experience. Unfortunately, for way too many students, organic chemistry becomes this um this right of passage, something you learned through sheer volume. cram it in and then poof, it's gone.

They get really focused on just making the mechanism work, you know, to get to the product they already know is coming instead of actually understanding the logic, the sort of underlying physics of why those electrons move where they do.

So, our mission today is basically to flip that script. We're doing a deep dive into a different way of thinking. It's a set of fundamental

kind of algorithmic priority rules. And the idea is the these rules can turn organic chemistry from just a huge pile of facts you have to memorize into something logical, something predictive. We want to give you the framework to actually propose plausible mechanisms from scratch, even for molecules you've never seen before.

Exactly. It's really about empowering you, the listener, to approach any problem in the molecular world with a bit more confidence. Now, some, you know, more advanced ways of teaching jump straight into the heavy math,

right? The quantum mechanics stuff.

Yeah. Talking about reactions using concepts like homo, highest occupied molecular orbital and Mo, lowest unoccupied molecular orbital. And while that's scientifically spot-on, it's often, let's be honest, pretty mathematically scary for introductory students.

Oh, definitely. You need a lot of physics background for that.

You do. So, these priority rules we're talking about today, they give you that necessary uh cognitive shortcut. They offer a more algorithmic way to think about those curly arrows, and it actually mirrors how experienced chemists think about problems day-to-day. It just lowers that barrier to entry quite a bit.

Okay. So, it makes it more accessible. What's the foundation? for these rules. What do we need to know first?

Okay, so to really get these rules, we need basically three quick concepts under our belt. First up, inductive effects. We sometimes write that as plus or minus I.

Okay,

think of this like a constant tug-of-war over electrons. It's based purely on electro negativity. How strongly an atom pulls electrons and it works through the single bonds, the sigma framework. The more electronegative atom just continuously pulls electron density. way.

Got it. Like a constant background pull. What's next?

Next are mesomeric effects or pi donation sometimes called resonance. Now if inductive effects are that quiet tug-of-war, mesomeric effects are more like a big flowing river of electrons.

Ah

yeah these involve moving electron pairs through multiple bonds or systems with lone pairs. The pi system

and these are generally more powerful.

Oh much more significant usually. And they can operate over longer distances within the molecule compared to inductive effects. Very important.

Okay. Inductive tug-of-war, mesomeric river. What's the third piece?

The third piece tackles a common um conceptual hang-up. It's about intermediates in a reaction. You often hear textbooks talk about stable intermediates.

Yeah. And that always felt weird to me. If it's stable, why does it keep reacting? Shouldn't the reaction just stop there?

Exactly. That's the confusion. So, we actually prefer the phrasing energetically accessible intermediates. The point of a mechanism isn't to make something so stable it stops reacting. The point is to find the pathway that requires the least energy overall to get from start to finish. Accessibility, not rock solid stability is the key.

Energetically accessible. I like that. It makes more sense for something that's part of a process.

Precisely.

Okay. Foundation laid. Let's get into the rules themselves. The level one mechanistic priority rules. You said there's a guiding idea.

Yes. The whole process is guided by a pretty simple almost self-evident summary. The electrons have to start somewhere electron-rich, move somewhere electron poor, and end somewhere good.

Rich to poor to good. Okay, let's break that down. Rule one, starting somewhere rich, right? Rule one, where do the electrons start? We need to find the most electron-rich site first. And there's a hierarchy, an order of priority.

Okay, what's top priority?

Top priority, a full negative charge, an anion. That's the most urgent spot because that extra electron density is inherently unstable. It wants to go somewhere.

Makes sense. What's next?

Second priority is an uncharged lone pair, you know, like the ones on a nitrogen or an oxygen atom. Still electron-rich, but not as desperate as a full negative charge.

Okay? Anion then lone pair. Then

third, you look for a multiple bond, a double or triple bond. Those pi electrons are relatively accessible. And finally, fourth, maybe a weak single bond, which might get involved in certain substitution reactions.

Got it. Negative charge lone pair, multiple bond, weak single bond. That feels intuitive. Okay. Rule two. Move somewhere electron poor. The destination Rule two, where do the electrons move? This is about finding the site most desperate for electrons. Again, there's a hierarchy.

What's the neediest site?

Highest priority goes to a positively charged carbon or a carbocation either. Think about it. A carbocation only has six valence electrons. It's electronically starving. A proton has zero. These are immediate targets.

Okay, that makes sense. C^{++} or H^+ . But hang on, what about something like a positively charged oxygen? or nitrogen. They have a plus sign. Shouldn't they be high priority, too?

Ah, that is probably the single biggest conceptual trap for students. It's a crucial distinction. You do not attack the positively charged heteroatom directly.

Why not? It's positive.

Because atoms like oxygen or nitrogen, even when they carry a formal positive charge, usually already have a full octet of electrons. They aren't actually electron deficient in the same way as C^{++} is.

Ah, okay. So, the positive charge doesn't mean it needs electrons in its outer shell.

Exactly. So, instead of attacking the positive heteroatom itself, you attack an adjacent atom, usually a proton or a carbon. That attack then allows the electron pair holding that atom to push back onto the positive heteroatom, neutralizing its charge and restoring its lone pair. It's a critical difference.

Wow. Okay, that's huge. It's not just about the charge symbol. It's about the underlying electron count. So C^{++} and H^+ are priority one. Attacking next to a positive heteroatom is priority two.

You got it. Priority two is attacking an atom attached to a positively charged heteroatom. Priority three would be more distant sites that can relay electron density towards a positive charge often through those mesomeric effects we mentioned.

Right. Like down a conjugated system.

Precisely. And then lower priorities four and five include attacking partially positive atoms δ^+ plus δ^+ carbons or hydrogens like in a carbonyl group or maybe further along an alternating pathway.

Okay. Rich source poor destination. Now rule three. End somewhere good.

The electrons have to land safely.

Rule three. the electron density ends up somewhere good. This is basically about not violating fundamental rules like the octet rule. Where the electrons end up depends on where they started and where they went.

So what counts as good?

Well, number one, the electron density is quenched by that starving C^{++} or H^+ . A new bond forms. Their valence shell is filled. Often the mechanism stops there.

Simple enough.

Number two, maybe the electrons break a pi bond. Typically pushing electron density on to a heteroatom. Think about attacking a carbonyl. The electrons attack the carbon and the pi bond breaks, putting a negative charge on the oxygen. That avoids giving carbon five bonds, right? Prevents octet violation.

And third, maybe the electrons break a sigma bond. Typically kicking off a heteroatom as a leaving group, like breaking a C-Cl bond, the chloride ion leaves. That electronegative heteroatom is a good place for that extra electron density to end up.

Okay. Start rich, move poor, and good. That covers the flow. You mentioned operational rules, too.

Yeah. Two quick operational checks. Rule four, charges must be balanced. This is the most common mistake. If you start neutral and use a neutral lone pair to make a new bond, the atom that donated the pair must now carry a positive charge. Always do that charge accounting.

Crucial bookkeeping. And rule five.

Rule five. This may not be the end of the process. Very often the first step creates a new reactive site. Maybe a new negative charge or positive charge. If so, you just loop back to rule one and apply by the rules again until everything is stable or quenched.

Right. It can be multi-step process. Apply, check, repeat if needed.

Exactly.

Okay, let's try this out. Can we walk through an example? Maybe something practical.

Absolutely. Let's use a real world pharmaceutical example. The synthesis of lidocaine. It's a common local anesthetic used a lot in dentistry. The reaction involves diamine reacting with a chloroamide derivative.

Okay. Two molecules, diamine and this chloro something.

Chloroamide derivative. Yeah. So, rule one, where do the electrons start? Where's the richest site.

Well, diamine has a nitrogen with a lone pair. The chloroamide has oxygen lone pairs, but also that chlorine-hydrogen nitrogen is generally a better donor than oxygen, right? Less electronegative. Good thinking though. We'll refine that with level two rules later. For level one, the lone pair on the nitrogen of diamine rule 1.2 is our best starting point. Now, rule two, where does it go? Where's the electrophilic site? The chloroamide thingy has a carbonyl carbon that's delta positive. And it has a carbon attached to the chlorine, also delta positive because chlorine pulls electrons, right? We have two potential delta plus sites, which is the better target.

Well, rule three mentioned breaking a sigma bond to a heteroatom as a good ending. Chlorine is a pretty good leaving group. So maybe attacking the carbon bonded to the chlorine lets us kick out the chloride ion.

Exactly. That's the spot. The lone pair on the diamine nitrogen, rule 1.2, attacks the partially positive carbon attached to the chlorine, rule 2.4. And following rule 3.3, Three, the electron density in the C-Cl bond gets pushed onto the chlorine, breaking the bond and making it leave as

text. Okay, new CN bond formed. Chloride ion kicked out. Now, rule four, check. The nitrogen started neutral, used its lone pair, so it must now be positive.

Correct. The nitrogen now has four bonds and a positive charge. The chlorine is a negative chloride ion. Charges balance plus one and one overall. Now, Now, rule five. Are we done? Are there reactive sites left?

So, we definitely have charges. The positive nitrogen and the negative chloride ion. So, no, we're not done.

Back to rule one. What's the most electron-rich site now?

The chloride ion. It has that full negative charge. Rule 1.1.

And where does it move? Rule two. It needs an electron poor site. It could be attracted to the positive nitrogen. But wait.

Ah, the trap. Don't attack the positive heteroatom directly. It probably has its oct.

Bingo. So, the chloride ion looks nearby. What's attached to that positive nitrogen?

Hydrogen atoms. Protons essentially made acidic by the positive nitrogen next door.

Right? So the text rule 1.1 attacks one of those adjacent protons. Rule 2.2 attacking H attached to a positive heteroatom. The electron pair in the NH bond then gets pushed back onto the nitrogen. Rule 3.3 sort of neutralizing charge on heteroatom

and that neutralizes the nitrogen uses up the chloride

and gives us the final lidocaine molecule.

Precisely reaction complete.

Wow. Okay. So just by following those steps rich, poor, good, balance, repeat. We navigated a real synthesis. That's pretty cool. It feels much less like guesswork.

And what's fascinating, as you said, is that these rules are basically a simplified way, an algorithm for estimating how the orbitals interact. The highest energy electrons, finding the lowest energy empty spot, LUMO,

right? It's like you don't need to be a mechanic to drive a car. You don't need the full quantum calculation for every arrow you draw. These rules let you navigate.

Exactly. But sometimes the road forks, sometimes rule one or rule two give you multiple options at the same priority level.

Ah, like two different atoms both have lone pairs or two different carbons are both delta positive. How do you choose them?

That's where we need the next level. Level two mechanistic selectivity rules. This is where the real chemical judgment comes in. These help you decide between sites that seem equally good based on level one.

Okay, so level one is can it happen? Level two is which happens better or faster.

Nicely put. Let's break down level two. First set selectivity for where the electrons start. If you have say two different atoms with lone pairs, both rule 1.2, which one is more likely to donate its electrons?

Okay, how do we decide?

The number one factor is electroactivity and it's often counterintuitive. The less electronegative atom is generally the more reactive nucleophile.

Wait, less electronegative means more reactive. Why?

Because it holds onto its electron pair less tightly. Think about the H halogens as nucleophiles. Iodide takes these way better than fluoride. Iodine is much bigger and much less electroactive than fluorine. That negative charge or lone pair is looser, more available to react.

Ah, okay. So, it connects back to that idea of stable versus reactive. Fluoride holds its electrons tight. So, it's stable but less reactive. Iodide holds them loosely, less stable maybe, but much more reactive.

Exactly right. It clarifies that charge alone isn't the whole story. How tightly that charge or lone pair is held matters hugely. A brilliant example is making paracetamol. or acetaminophen the pain reliever.

That's the one. The starting molecule has both an amine group nitrogen with a lone pair and an alcohol group oxygen with lone pairs. Both are rule 1.2 sites in the level one rules.

Right? Nitrogen lone pair oxygen lone pair. Level one doesn't choose between them.

But level two does. Nitrogen is less electroactive than oxygen. Therefore, its lone pair is held less tightly. It's more available. It's more nucleophilic. So the amine nitrogen is much more reactive than the alcohol oxygen.

So the rule predict the nitrogen will react first.

Absolutely. And that's exactly what happens in the synthesis. Knowing that immediately tells you the correct first step. It's powerful.

Okay. Electro negativity is key for choosing the starting point. What else? For level two starting points.

We also have to consider the effects of resonance. If a lone pair or charge is heavily involved in resonance, it's spread out delocalized. That makes it more stable and therefore less reactive.

Okay. So resonance stabilization decreases reactivity.

Correct. And finally, steric, how bulky things are. A really big bulky molecule with a lone pair might be a poor nucleophile attacking carbon because it just can't physically get close enough.

So, it might do something else instead.

Often, yes. Bulky electron-rich things are often better bases. They'll attack a small accessible proton H^+ instead of a crowded carbon atom. Steric influences what kind of reaction happens too.

Fascinating. Okay, that's choosing the start. What about the level two rules for where the electrons move to? Choosing between two potential electrophiles.

Similar principles apply. First, electron deficiency. We use inductive and mesomeric effects again to figure out which delta plus carbon site is most electron poor. For instance, think about an amide versus an ester. Both have carbonyls. CO,

right? Both delta plus carbon carbons.

But in the amide the nitrogen's lone pair can resonate, pushing electron density towards the carbonyl carbon via a mesomeric effect that makes the amide carbonyl carbon less electron poor, less electrophilic than the ester carbonyl carbon where the oxygen is mostly just pulling electrons away inductively.

So the ester carbonyl is more delta positive more reactive

significantly more reactive yes that's why proteins use stabilamide linkages in their backbones nature wants stability there not high reactivity like you'd get with ester links

that makes so much sense biologically what about sterics for the destination

sterics plays a huge role here too bulkiness around the target site matters think about reactions with epoxides those three membered rings with an oxygen. Attack usually happens at the carbon atom that has fewer other groups attached. The less substituted carbon

because it's easier to get to.

Exactly. It's less crowded, easier for the nucleophile to approach. Often electronic effects and steric effects reinforce each other pointing to the same site.

So level one gives a basic pathway. Level two helps you choose when there are forks in the road using electronegativity, resonance, and sterics.

That's the core idea. And what it all boils down to is moving away from just memorizing reaction after reaction. Instead, you gain this coherent framework. You start seeing organic chemistry as predictive. It actually, I think, reduces the real burden of learning the subject. Even though it seems like you're learning rules,

it's like learning phonics to read instead of memorizing whole words. You get the fundamental building blocks.

And this framework, it's robust, right? Even when things get weird.

Absolutely. Take something classic like SN1 versus SN2 reactions. Sometimes the outcome seems to diverge from a simple prediction. But these rules give you the baseline. You can then ask why the specific molecule behaves differently. Usually it relates back to the stability or energetic accessibility of the intermediates involved like carbocations and SN1. The rules provide the reference point for understanding the exceptions.

That's really powerful. So thinking bigger picture. How does a chemist use this kind of thinking for something really complex?

Well, think about synthesizing a really complex natural product. A famous example is the anti-cancer drug paclitaxel. Taxel also known as paclitaxel. It's an incredibly complicated molecule.

Right. Originally found in utrees.

Exactly. The starting material they could get more easily from you needles called 10 detal backin is already complex and it has multiple alcohol groups O groups. To turn it into taxol, chemists needed to modify one specific alcohol group while leaving the others untouched.

How on earth do you do that selectively on such a big molecule?

You use these principles, especially level two selectivity. You look at the 3D structure of the molecule. You apply the rules of sterics, which alcohol group is sticking out exposed. Which ones are buried deep inside the structure sterically hindered?

Ah, so you figure out which one is easiest to get to

precisely based on that steric accessibility which is predicted by understanding the molecule shape and applying these ideas. They figure out a strategy. Often they'll chemically protect the most reactive, most exposed group temporarily, do reactions on the less accessible sites and then remove the protecting group at the end. So it's all about predicting reactivity based on structure and electronics using these fundamental ideas.

It is it's first principles thinking applied even to incredibly complex life-saving molecules. That's the real power of moving beyond memorization.

PODCAST VERSION 2

USER PROMPT:

No prompt given

AI OUTPUT

From Cookbook to Logic: Mastering Organic Chemistry Mechanisms with 5 Priority Rules

So, if you've ever stepped into an organic chemistry 101 lecture, you probably know the reputation. It's uh pretty infamous, right?

Oh, absolutely. Often seen as this like necessary evil, a right of passage you just have to grind through, memorize a ton, and then well, forget most of it.

Exactly. And the way it's often taught, well, when you hit those complex reaction mechanisms, it feels like critical thinking just goes out the window.

That's a really common observation. Students often stop trying to follow the actual electron flow, the curly arrows, and just start, you know, trying to force the atoms together to make the product they think they're supposed to get.

I definitely remember that feeling, like trying to memorize a cookbook with a thousand slightly different recipes. The sheer volume of stuff can push actual problem solving aside.

And that's precisely why we're diving into this today. We're exploring a completely different approach, a logical um rule-based framework. It's designed specifically to replace that rote learning right? A set of mechanistic priority rules. So, our goal here is to really unpack these rules for you, show how they let you look at pretty much any set of starting materials, even ones

you haven't seen before, and logically predict a plausible reaction mechanism. All from first principles.

It's about shifting from being a memorizer to becoming a well, a mechanistic thinker.

Less flash card flipping, more thinking.

Precisely. It moves away from specific named reactions and focuses on a versal logic. Now, don't get me wrong, there are very rigorous methods like using molecular orbitals, homolumo interactions.

Yeah, but that set can be pretty heavy going, especially if you haven't got a strong physical chemistry background.

It really requires significant prior learning. These rules though, they offer a kind of algorithmic shortcut. They break down what seems like a complex abstract chemical puzzle into maybe five smaller solvable steps.

Okay, I like the sound of that. So, before we jump into the rules themselves, we probably need to lay some ground work, right? Get some basic concepts straight.

Absolutely. Foundational thinking. We need to be on the same page with terminology. Let's start with electronic effects. First up, inductive effects.

Okay.

These are all about electro negativity differences between atoms. And they operate through the sigma bonds, the single bonds, kind of the baseline electronic landscape if you will. We use plus I to denote donating or withdrawing through this effect.

Got it. Sigma bonds, electro negativity. Then there's the other type. Resonance.

Well, yes, resonance is the common term, but the sources we looked at often prefer misome effects. It's essentially the movement of electron pairs through the pi system. Double bonds, triple bonds, lone pairs next to pi systems.

And why the preference for misome?

It just provides a nice parallel nomenclature. You have plus ISA for inductive and you have plus ism for misome. It cleanly separates donating or withdrawing density via the sigma framework versus the pi framework. And crucially, these misameric effects, the plusim ones, are often much much stronger than the inductive ones. They can really dominate the reactivity.

Ah okay, that unification makes sense. Now another key idea maybe more of a mindset shift is about reaction intermediates.

Yes, this is a huge stumbling block. Students here are stable intermediate and they think

oh it's stable so it stops there. It's unreactive. I definitely made that mistake.

Exactly. It's a massive misconception. So to combat this we need to shift our language. We talk about the principle of energetically accessible intermediates.

Energetically accessible. Okay. So what does that really mean in practice?

It means we stop focusing on how stable the intermediate is in isolation and start thinking about the energy barrier, the transition state energy needed to get to the next step. Is the pathway forward energetically easy to reach?

Right? So it's not about getting stuck in an energy well. It's about finding the lowest energy path, the easiest route for the reaction to keep progressing.

Precisely. And this whole system, all five rules we're about to get into. They essentially boil down to one simple mantra, a guiding principle.

Okay, Leonas.

The electrons must start somewhere electron rich, move somewhere electron poor, and end somewhere good.

Rich to poor to good. Sounds simple enough. Let's break down how we actually make that happen with the curly arrows. Rule number one,

PR1. The electrons start somewhere electron rich. This rule gives you a priority list for where to draw the tail of your first curly arrow.

Okay. What's top priority?

Highest priority is a formal negative charge. That's your most electron rich spot. If you don't have that, look for an uncharged lone pair like on a nitrogen or oxygen.

Makes sense. Negative charge first, then lone pair. What's next?

After lone pairs, you look for a multiple bond, a pi bond basically. And then lowest priority is a particularly weak single bond, though that's less common in introductory examples.

Got it. Negative charge, lone pair, pi bond, weak sigma bond. That covers where to start. Now, where do they go PR2 the electrons move somewhere electron poor

right the arrowhead top priority destination a positively charged carbon atom a carbocation or H plus it's a proton

why those first

well they have an urgent formal need for electrons that carbocation only has six electrons around it instead of eight it's electron deficient in a big way

okay urgent needs satisfied first what's the next level down this is where it gets tricky isn't it

yes this is the classic trap the next priority target is an atom attached to our positively charged heteroatom. So think carbon or hydrogen next door to an O plus or N plus.

Critically, you're saying attached to it, not the positive heteroatom itself.

Absolutely critical. You cannot point the arrow directly at that N plus or O plus Y because it already has a full octet of electrons. Even with the positive charge, attacking it directly would instantly give it 10 electrons,

which is a huge no no in chemistry. Breaks the octet rule.

Exactly. Chemically unthinkable for these elements. So you must attack the adjacent atom, the neighbor. This allows the electron pair already bonding to the heteroatom to move back onto it, neutralizing that positive charge.

Okay. Attack the neighbor to relieve the charge on the heteroatom.

Oh, got it.

And the lowest priority for PR2.

That's attacking a carbon or hydrogen that isn't fully positive, but just partially positive δ^+ δ^+ usually because it's attached to an electronegative atom like oxygen or a halogen.

All right, which leads perfectly into the next rule because attacking a δ^+ site creates a new problem.

It does indeed. PR3, the electron density must end up somewhere good. This rule is necessary only if your PR2 step involved attacking one of those δ^+ centers.

Why only then?

Because if you attacked a δ^+ atom, there wasn't a formal positive charge waiting to accept the incoming electrons. So if you just stopped after forming the new bond, the atom you attacked would exceed its octet.

Ah, okay. So PR3 is about immediately fixing a potential octet violation. caused by attacking a neutral but polarized atom. Where does the density typically go?

Usually, it has to push electrons onto an electronegative heteroatom nearby. This often happens by breaking a π bond like in a CO group. The electrons move on to the oxygen as a lone pair. Or you might break a σ bond entirely, kicking off a leaving group like a chloride ion.

Okay, start rich. PR1, move poor, PR2, and somewhere good if needed, PR3. What's next? Bookkeeping.

Pretty much PR4. charges must be balanced. This sounds obvious, but it's so easy to mess up.

What's the common mistake?

Forgetting to add a positive charge. When a neutral atom, like nitrogen, uses its lone pair to form a new bond, the electrons start neutral, they form a bond, so the atom that donated them must become positive. Always check the overall charge before and after each step.

Good check. And finally,

PR5. This may not be the end of the process. Very often, the first mechanistic step generates a new reactive species. usually something charged. If that's the case, you have to iterate.

Go right back to rule one. Look at your new intermediate. Find the most electronrich spot now. Find the most electron poor spot now. Apply the rules again. Keep going until you form a stable neutral product or run out of reasonable steps.

Okay, let's see that iteration in action. You mentioned using cellbutimal, the asthma drug, as an example.

Yes, a key step in its synthesis is perfect. We start with a molecule that has a primary amine group, an NH_2 , and we react it with another piece that has a carbon attached to a chlorine atom. Sort of like an alkyl chloride derivative.

Right. Step one, PR1.

The most electron-rich spot is the lone pair on the nitrogen of the amine. That's PR1 priority 2 uncharged lone pair.

Okay. Where does it go? PR2.

It attacks the carbon attached to the chlorine. That carbon is electron core. It's delta plus because chlorine is electronegative. So that's PR2 priority 3 attacking a delta plus carbon.

Ah, and because we attacked a Delta plus we need PR3 immediately.

Exactly. To avoid breaking the octet on carbon, the C-Cl bond has to break and those electrons move onto the chlorine atom,

creating a chloride ion Cl^- . And what's left with the main molecule?

Well, the nitrogen formed a new bond using its lone pair. So, PR4 tells us the nitrogen now has a positive formal charge N^+ . And we also have the Cl^- around.

So, we have a charged intermediate N^+ plus and Cl^- .

PR5. Iterate. Go back to rule one. What's the most electron rich thing. Now the chloride ion Cl^- it has a negative charge so that's PR1 top priority.

Perfect. Now PR2 where does it go? It's attracted to the positive nitrogen but remember it can't attack the N^+ directly. Octet rule violation. It has to attack an adjacent atom.

Precisely. In this case the easiest adjacent atom to attack is one of the hydrogens attached to the N^+ . The Cl^- acts as a base here.

So the Cl^- deprotonates.

Yep. It attacks the H. The electrons that were in the NH bond then move back onto the nitrogen atom

ah which neutralizes the positive charge on the nitrogen and we're left with a final stable organic product plus HCl essentially

exactly a clean two-step mechanism derived entirely by following the priority rules including the iteration

that example also neatly shows the choice that sometimes happens in PR2 right the chloride acted as a base attacking hydrogen not as a nucleophile attacking carbon

correct the electron-rich species often has that choice attack a carbon be a nucleophile or attack a proton, be a base. Now, for beginners, we often just tell them which role or region plays to simplify things

reduces the cognitive load as you said,

right? But understanding why something chooses to be a base or a nucleophile factors like sterics, electronics, that's where we need a bit more sophistication. That leads us towards level two rules.

Before we get to level two though, let's just connect this back. These rules feel intuitive, but what's the deeper chemical principle making them work? Is it related to those orbitals you mentioned earlier.

It absolutely is. Think of these priority rule as a very useful um practical shortcut. What they're really doing in a rough way is estimating the best possible overlap between the highest occupied molecular orbital the homo

where the electron pair starts

and the lowest unoccupied molecular orbital the lomo

where the electron pair is going.

Exactly. The reaction happens most easily when the highest energy electrons can flow into the lowest energy empty spot. So when we draw an arrow from say a lone pair on an oxygen which is in a high energy non-bonding orbital the homo to an empty orbital on a carbocation the lomo that arrow represents the formation of a new lower energy sigma bonding orbital.

So the rules are basically guiding us to the most energetically favorable homolumo interaction without needing to calculate the orbitals themselves.

That's a great way to put it. You don't need quantum mechanics to follow the rules, but it's reassuring to know they're grounded in sound physical principles.

Okay, good. Now complexity. What happens in bigger molecules where maybe you have say two different lone pairs or two different delta plus carbons, how do you choose?

Ah yes, now we enter the realm of selectivity. This is where you might have multiple sites that fall under the same priority level in PR1 or PR2. We need level two mechanistic selectivity rules to make the right choice.

Makes sense. Let's start with selectivity rule one. Where the electrons start? If you have two potential starting points, how do you pick?

Okay, here's a slightly counter-intuitive one related to electro negativity. If you're comparing two different atoms as electron donors say comparing iodide I versus fluoride F acting as nucleophiles which is better?

Fluorine is more electronegative. So maybe it holds electrons better. So iodide is better at giving them away?

Exactly. It seems backward at first but the curly arrow means the electrons are leaving. The less electronegative atom holds its electrons less tightly making it a better donor. So I is actually a much better nucleophile than F. Interesting. What else affects the starting point choice?

Mesomeric effects again. If a lone pair is heavily involved in resonance or mesomeric stabilization, it's delocalized. It's spread out and more stable,

making it less available to react.

Precisely. That's why an amide nitrogen is a much poorer nucleophile than an amine nitrogen. In the amide, that lone pair is busy being shared with the carbonyl oxygen through the plus-m effect. It's less available to attack something else.

And sterics both iness

huge factor especially for the nucleophile versus base decision.

Generally less bulky reagents make better nucleophiles because they can physically reach the carbon atom they need to attack

whereas a big bulky electronrich thing

might find it really hard to get close to an equally bulky carbon center but a small proton H^+ on the periphery of a molecule that's much easier to reach. So very bulky bases like potassium t -butoxide are great at removing protons but terrible as nucleophiles. Steric bulk favors basicity.

Okay, so electronegativity less a better donor. Deoxygenation reduces reactivity and steric bulk favors specificity. Now for the destination selectivity rule two where the electrons move to. What if you have multiple electron poor sites?

Here it's mostly about figuring out which site is most electron poor which has the largest delta plus charge

and that depends on the surrounding atoms

entirely. Think about carbonyl groups again. CO is the carbon in a ketone more or less delta positive than the carbon in an aldehyde. It depends on what else is attached. Alkyl groups are slightly electron donating plus effect. So they reduce the delta positive character. Aldehydes only have one alkyl group and one H while ketones have two.

So the aldehyde carbon is slightly more delta positive making it generally more reactive towards nucleophiles than a ketone carbon.

Correct. It's all about the subtle push and pull of electrons governed by those inductive and mesomeric effects.

And you mentioned a really cool connection here to biology.

Yes. It's fascinating. Why are proteins the workhorses of life built from amide linkages joining amino acids, not ester linkages?

Okay. Why? Based on what we just said, the amide nitrogen has a plus m effect.

Exactly. That nitrogen pushes electron density onto the amide carbonyl carbon, making it significantly less delta positive, less electron poor than the carbon in an ester group, which only has an oxygen's inductive effect and weaker plus M effect pulling density away. So the amide bond is just less reactive,

much less reactive towards hydrolysis breakdown by water. And that stability is essential for proteins to function and persist in the aqueous environment inside cells. Life literally relies on that subtle difference in reactivity dictated by these electronic effects.

That's incredible. So wrapping this up, the real power of these priority rules, it's about providing a mental framework, isn't it?

Absolutely. It lowers the cognitive load instead of memorizing hundreds of reactions, you learn one logical process. We consistently see students respond really positively. They appreciate learning a method they can apply rather than just specific examples.

It transforms you from just a memorizer into someone who can actually think mechanistically.

That's the goal. And here's a final thought to leave you with. These rules predict the most plausible pathway. Often that's something like in Texen 2 reaction nucleophile attacks leaving group leaves all driven by the priorities.

But reactions don't always follow the most obvious path.

Exactly. Sometimes conditions favor a different route like in Texendall one mechanism where the leaving group leaves first to form a carbocation, especially if the nucleophile is weak or the carbocation is very stable.

So the rules might predict SN2, but SN1 happens,

right? And the real beauty of understanding the priority rules is that they give you the baseline. They show you the default pathway. So when a reaction deviates from that path, the rules give you the perfect framework to ask why. What specific condition poor nucleophile stable intermediate solvent effect cause this reaction to subvert the standard priorities.

So understanding the expected helps you analyze the unexpected

precisely and developing that ability to reflect on why a mechanism follows a particular sometimes non-obvious path. That analytical skill is arguably the true endgame of learning organic reaction mechanisms.

PODCAST VERSION 3

USER PROMPT:

Explain to a student why we need to develop priority and selectivity rules to understand organic mechanisms, and then explain the basics of the priority and selectivity rules and how they work in practice to predict organic reactions.

AI OUTPUT

From Rote Memorization to Operating System: The Priority Rules That Unravel Organic Chemistry Mechanisms

Okay, let's talk about organic chemistry or chem 101. If you've been through it, you probably remember that feeling. Maybe dread.

Oh, definitely. It often feels less like learning and more like uh surviving a massive info dump.

Exactly. Like this huge mountain of reactions, weird named mechanisms, endless facts, all just stuff to cram in, endure, and then, let's be honest, forget immediately after the exam.

That's such a common experience. I remember students just trying to make the errors work to get the product, not really thinking about why the electrons move that way.

Yeah. Like reverse engineering it without understanding the cause and effect.

Precisely. And you know, the way it's traditionally taught kind of encourages that you race through so many different reaction types without really nailing down the fundamentals first.

Right. So what if we could flip that?

Yeah.

Instead of a library of facts to memorize, what if organic chemistry could be more like an operating system, something logical?

That's the idea behind the source material we're looking at today. It proposes this really um elegant alternative, a simple set of priority rules.

Okay, priority rules. The goal here, as I understand it, is to learn how to think mechanistically before you even get bogged down in specific named reactions.

Exactly. It's about empowering you, the listener, to look at molecules and predict how they might react based on first principles, even complex stuff, maybe reactions you haven't explicitly been taught.

That sounds incredibly useful because organic chemistry isn't just an academic. Is it? It's well, it's everywhere.

It really is. It's the basis for drug discovery, material science, molecular machines, you name it. But if we're just testing recall like a lot of current assessments seem to do, we're not building that deeper critical thinking needed for real innovation.

So instead of learning reaction recipes, mix A and B get C. We need a more um algorithmic way, a system for figuring out those curly arrows, the electron movements.

Yes, these priority rules offer that framework. They give you a structured way to find the reactive spots in molecules and predict where the electrons are likely to go. It gives you a starting point, a way into the problem.

And the core idea behind all these mechanisms, it's actually pretty simple when you boil it down.

Deceptively simple, really. At its heart, any organic reaction mechanism, no matter how complicated it looks, follows three basic conditions. You need electrons starting somewhere electronrich.

Okay? Like a source

moving to somewhere electron pool

that destination

and ending up somewhere well somewhere good or stable. We're going to formalize this with five specific priority rules that you apply in order.

All right, let's get into it. Level one priority rules. Rule one, where did the electron start? Finding that electronrich site.

This is the absolute highest priority starting point for drawing that first curly arrow. And there's a fixed hierarchy.

Okay, number one.

Number one is a negative charge. Can't get much more electronrich than having an extra electron pair sitting there. All right, it's basically shouting, "Use me."

Makes sense. It's next.

Number two is an uncharged lone pair. Think of the electrons on a nitrogen in ammonia or an oxygen in water. Still very available, but maybe not quite as uh pushy as a full negative charge.

Okay, negative charge then lone pair. What follows those? We get into bonds now. Right.

Right. Number three is a multiple bond, which we call a pi system. Think double bonds, triple bonds. Those pi electrons are relatively high energy. more exposed, easier to get involved in reactions compared to single bonds.

And the last resort,

number four, the lowest priority starting point is a weak single bond, a sigma bond. Usually, this involves bonds to large, easily polarized atoms like iodine or maybe strained ring systems. You typically only break these strong sigma bonds if there are no better options available.

So, it's a strict order. If you have both a negative charge and a lone pair on the same molecule. The negative charge always dictates where the first arrow starts.

Always. Rule one is absolute in its sequence. Negative charge first, then lone pair, then pi bond, then sigma bond.

Got it. So, rule one gives us the starting block. Now, rule two, where do those electrons go? The electron pore site, the target.

Yep. Where the arrow needs to point. And again, there's a priority order. Highest priority target.

I'm guessing a positive charge.

Exactly. Number one is a positively charged carbon uh a carbium ion or just a proton H^+ . These species are seriously electronic defic. A carbium ion only has six electrons around it. H^+ has none. They desperately need electrons to get a full valence shell.

Okay, that seems straightforward. But the second priority target, this is where you mentioned it gets interesting and potentially confusing. It's an atom attached to a positively charged heteroatom.

Correct. Like a hydrogen attached to a positively charged oxygen or nitrogen.

Okay, hold on. Why not attack the positive heteroatom directly? Isn't positive the most attractive target according to priority one here? This feels like a prediction.

Ah, that's a fantastic question because it hits on a fundamental principle.

Yeah,

the octet rule.

Remember atoms like oxygen or nitrogen want eight valence electrons. Even when they have a positive formal charge like an H_3O^+ , that oxygen already has a full octet.

Oh, right. Okay.

So, if your electron pair, your nucleophile trying to attack that oxygen directly, you'd be trying to force 10 electrons onto it. That's a huge no no in chemistry. Violates the octet rule big time.

So, the attack has to happen nearby. to sort of relieve the situation without breaking the octet rule.

Precisely. The arrow attacks an adjacent atom, usually a hydrogen or sometimes a carbon. This allows the electrons in the bond connecting that atom to the positive heteroatom to shift back onto the heteroatom.

Ah, so it neutralizes the positive charge by giving the heteroatom its lone pair back, satisfying the octet. Clever.

It's a crucial distinction for getting mechanisms right. So after positively charged C or H and atoms next to positive hetero. What's the next level of electron porite?

That would be number three. A partially positive carbon or hydrogen. The delta plus ones,

right? These occur when carbon or hydrogen is bonded to something much more electronegative like oxygen, nitrogen or a halogen like chlorine. That atom pulls electron density away leaving the C or H slightly electron poor or delta positive.

Okay. So we have a start. Rule one, a target rule two. Now rule three. Where does the electron density actually end up? The somewhere good rule. You called it. This stops us breaking that octet rule.

Exactly. When the new bond forms, something often has to give way. There are three main outcomes for where that electron density goes.

First one,

the simplest is if the target was a C⁺⁺ or H⁺ rule 2.1, the new bond just forms, quenching the charge. Mechanism might stop there. Stable bond formed. Done.

But if we attacked a delta plus atom, like in a carbonyl group,

right? If you attack a delta plus carbon that's part of a double bond, to a heteroatom like CO the electron density often shifts onto that heteroatom by breaking the pi bond so the CO becomes CO minus that negative charge on oxygen is pretty stable that's outcome number two

third way

the third way is breaking a single bond a sigma bond usually to a heteroatom this is what we call a leaving group the incoming electrons force the electrons in that sigma bond to leave entirely with the departing atom or group think of Cl leaving

okay so the electrons push out another group makes sense that covers the main flow what about the supporting rules Rule four and five.

Rule four is charge balance. It's basically just careful accounting. If a neutral thing like an amine nitrogen using its lone pair forms a bond, it must gain a positive charge.

And conversely, if a positive atom gets its lone pair back,

it loses the positive charge. Forgetting these charges is probably the single most common mistake students make when drawing mechanisms. Got to balance the books.

And rule five, iteration.

Yeah, this just acknowledges that reactions often happen in multiple steps. If your first mechanistic step following rules 1 to four, creates a new reactive site, maybe that minus we mentioned, or a new positive charge, well, that new site might be the starting point for the next step.

So, you just loop back, apply rule one again to the new situation.

Exactly. You keep applying the rule sequentially. Find the best electron source, find the best target, see where the electrons end up, balance charge until you reach a stable product or run out of reactive species. The mechanism unfolds step by step following these priorities.

Okay. Theory is great but let's see it work. You mentioned an example a key step in making salbutamol the asthma drug.

Yes. Let's take tertiary budolamine reacting with a molecule that has an epoxide and a chloromethyl group. We'll focus on the reaction at the chloromethyl group C.

All right. Step one rule one find the most electronrich site.

We have the nitrogen lone pair on the budolamine and maybe oxygen lone pairs if there were other groups but nitrogen is less electronegative than oxygen. That makes its lone pair a better donor, more available. So the N lone pair is our starting point.

Okay. Rule two, where does that nitrogen lone pair attack?

It targets the carbon attached to the chlorine. The chlorine is electronegative, pulling electron density away, making that carbon partially positive, delta plus. That's our electron poor site. Rule 2.3.

Got it. Rule three. What happens as the N attacks the C? Where do electrons end up?

To avoid giving that carbon five bonds, breaking the octet rule, the C-Cl bond has to break. The pair of electrons in that sigma bond moves entirely onto the chlorine atom. This satisfies rule 3.3 breaking a sigma bond to a heteroatom. Chlorine leaves as Cl^- .

Okay, so the nitrogen bonds to the carbon and the chlorine pops off. Now rule four, charge balance.

Crucial step. The nitrogen was neutral, used its lone pair to make a bond, so it now gets a positive charge. The chlorine atom took the electrons from the C-Cl bond. So it becomes a negatively charged chloride ion Cl^- .

Right? So we've formed one bond, broken another, and And we have new charges N plus and Cl^- . Is the reaction done?

Not quite stable yet. We have charged species.

So rule five, iterate. Go back to rule one. What's the most electronrich site now?

Uh the chloride ion Cl^- . It's got a full negative charge that beats the neutral lone pair on the nitrogen. Right. Wait, no. The nitrogen's positive now. So yes, Cl^- is the source.

Exactly. Cl is our new electron source. Rule 1.1. Now rule two. Where does it attack? What's the most electron poor site.

Well, we have that positively charged nitrogen, but we can't attack it directly. Octet rule. So, we attack something next to it, like one of the hydrogens's attached to the nitrogen.

Perfect. That hydrogen is very electron poor. Rule 2.2 because it's attached to the N plus. So, so the Cl attacks the H+.

By rule three, what's the consequence?

The electrons in the NH bond, which are being pulled towards the N plus anyway, shift entirely back onto the nitrogen atom.

Ah, giving the nitrogen its lone pair back.

Yeah.

And rule four,

the positive charge on the nitrogen disappears because it regained its lone pair. The Cl and H+ form neutral HCl and the main organic molecule is now neutral and stable. Mechanism complete.

Wow. Okay. Following those steps really maps it out. But you mentioned sometimes the rules give you options, right? Like what if there were two different lone pairs that looked equally good according to rule one.

Exactly. That's where level one rules might give you say two potential starting points of the same priority or two potential targets. How do you choose the better one? That's what level two the selectivity rules are for.

So level two adds nuance. It helps us break ties when level one presents multiple options.

Precisely. It brings in factors like electro negativity differences, how bulky molecules are sterically, and the stability of intermediates.

Let's look at selectivity rule one. Choosing the starting point when level one gives options. You said something counterintuitive here.

Yeah, this one surprises people. When comparing atoms in the same row, like N, V, S, F or the halides down a group, the electrogonic atom is often the better electron donor.

Wait, less electrogonic is better? Why wouldn't more electrogonic atoms want to hold electrons less?

They hold electrons tighter. Electro negativity means electron hogging. Fluorine holds its lone pairs very close to its nucleus. Iodine being much larger and less electrogonic has electron clouds that are more diffuse further out, more polarizable. This makes it easier for iodine's electrons to reach out and overlap with the orbitals of the atom it's attacking to form a new bond. So better orbital overlap from the less electrogonic more polarizable atom makes a better donor.

Okay. Example for halide ions as nucleophiles reactivity goes iodine bromine chlorine fluorine. I is the best F is the worst. Generally for neutral lone pairs in the same row nitrogen is usually a better donor than oxygen which is better than fluorine.

Okay. Electronegativity. What else does selectivity rule one consider? Steric effects.

Yes. Bulkiness. This introduces a key distinction. Nucleophile versus base. A nucleophile attacks a carbon atom. A base attacks a proton H^+ .

And size matters here

immensely. A small unhindered electron donor can easily approach and attack a carbon atom. Even if that carbon has other groups around it, it acts as a good nucleophile.

What about a big bulky donor.

A big bulky donor might find it hard to squeeze in to reach the carbon. It's sterically hindered. But attacking a small exposed proton on the edge of a molecule, that's much easier. So bulky electron-rich species often act as bases rather than nucleophiles. It's the path of least resistance.

Got it. So selectivity rule one helps pick the starting arrow based on electronics and size. What about selectivity rule two? Picking the target when there's a choice.

Again, it involves electronics and steric effects. We might have two different δ^+ carbons. How do we choose? We look at how δ^+ they are.

Can you give an example?

Compare the carbonyl carbon in an amide like in proteins versus an ester. Both have a CO group. So both carbons are δ^+ . But which is more δ^+ , more reactive?

H not sure.

It's the ester. In the amide, the nitrogen atom next to the CO uses its lone pair to donate electron density back towards the carbonyl carbon. It's called a resonance or mesomeric effect. This makes the amide carbon less electron-poor, less reactive.

Ah, so the nitrogen sort of counteracts the oxygen's pull.

Exactly. The oxygen next to the ester carbonyl doesn't donate back nearly as effectively. So esters are much more reactive towards nucleophiles than amides

and that's why proteins use amides they need to be stable

precisely they need to survive in water without falling apart amide bonds provide that stability esters wouldn't last it's a beautiful piece of biochemical design based on fundamental reactivity

okay that's the electronic part about steric effects for the target you mentioned an analogy the perfect bar

right the epoxide example an epoxide is a three-membered ring with oxygen when a nucleophile attacks to open the ring it usually has a choice between two carbons. Often one carbon is more substituted bulkier than the other.

So where does it attack?

It almost always attacks the less substituted carbon. And the beauty is both steric and electronics often point the same way. The less substituted carbon is easier to physically approach. Less crowded like the less crowded end of the bar. And often due to subtle electronic reasons, it's also slightly more electron deficient. The beer is cheaper there.

Okay. Less crowded and cheaper beer. Easy choice. attack the less substituted carbon.

That's the gist of it for epoxide opening under basic neutral conditions. Steric and electronics reinforce each other.

Right. One more selectivity rule. Rule three, where the electrons end up. This seemed related to intermediate stability.

Yes, it's about ensuring the reaction proceeds through the most stable or energetically accessible intermediate. The source material prefers that term over the more technical Hammond postulate.

Why avoid Hammond?

It sometimes confuses students. They hear stable intermediate and think it means the intermediate is unreactive, which isn't the point. Energetically accessible focuses on the fact that the pathway to forming that intermediate is lower in energy, meaning the reaction is faster.

So, the reaction takes the easiest route, the one with the most stable checkpoint along the way.

Pretty much think about adding something like HBr to an alkene. The first step is the alkene's pi bond attacking the H⁺. This forms a carbocation intermediate before the Br⁻. There usually two possible carbocations you could form, right? Primary versus secondary or secondary versus tertiary.

Exactly. And the rules predict you'll form the more stable carbocation intermediate. Tertiary is more stable than secondary, which is more stable than primary. Why? Because that intermediate is lower in energy, easier to form. That dictates which product you ultimately get.

So, putting it all together, these level one and level two rules give you a powerful toolkit. It's less about memorizing hundreds of reactions

and much more about understanding the fundamental principles of electron movement, charge, stability, and shape. It lets you work out how molecules are likely to behave just by looking at them. You develop a chemical intuition, a feel for it.

That's like learning the grammar of chemistry, not just vocabulary lists.

That's a great analogy. If curly arrows are the language, these rules are like the phonics or the basic grammar. They give you the building blocks to become fluent, to read and write chemical reactions with confidence.

So, the big takeaway for you listening is that you don't have to feel overwhelmed by organic chemistry. There's a logical system underneath it all.

Absolutely. These rules provide a step-by-step method to tackle mechanisms, even ones you've never encountered before.

Okay, final provocative thought time. We've built up this logical system. The level one rules, for instance, generally predict a mechanism called SN2 attack and leaving group departure happening at the same time.

Right? That's the standard outcome for many simple cases based on the priority rules.

But Chemistry loves exceptions. There's another common mechanism SN1 where the leaving group leaves first forming a carbocation and then the nucleophile attacks. The steps are in a different order.

Correct? And this is where the rules become even more powerful. By understanding the standard prediction from the rules, you can start to analyze why certain molecules or conditions lead to a different outcome like SN1.

So the rules help you understand the exceptions too.

Exactly. They give you the baseline. When reality deviates, you can ask why did the leaving group leave first here? What made that carbocation intermediate so accessible even though rule one didn't immediately point that way? So here's something for you to ponder. Why might the order of steps change if your electron source, your nucleophile isn't very strong and the carbocation that would form is particularly stable? Go look it up. It all loops back to those core principles of stability and reactivity.

FINAL TRANSCRIPT OF PODCAST WITH TIMESTAMPS

1

00:00:00,090 --> 00:00:03,870

[Auto-generated transcript. Edits may have been applied for clarity.]

Okay. Let's talk about organic chemistry. Org chem 101.

2

00:00:04,230 --> 00:00:08,460

If you've been through it, you probably remember that feeling. Maybe dread.

3

00:00:08,520 --> 00:00:14,580

Oh, definitely. It often feels less like learning and more like, uh, surviving a massive info dump.

4

00:00:14,790 --> 00:00:24,530

Exactly like this huge mountain of reactions, weird named mechanisms, endless facts, all just stuff to cram in, endure.

5

00:00:24,540 --> 00:00:28,020

And then, let's be honest, forget immediately after the exam.

6

00:00:28,440 --> 00:00:29,909

That's such a common experience.

7

00:00:29,910 --> 00:00:35,730

I remember students just trying to make the arrows work to get the product, not really thinking about why the electrons move that way.

8

00:00:35,760 --> 00:00:38,850

Yeah, like reverse engineering it without understanding the cause and effect.

9

00:00:38,850 --> 00:00:43,319

Precisely. And, you know, the way it's traditionally taught kind of encourages that you race through so many

10

00:00:43,320 --> 00:00:46,860

different reaction types without really nailing down the fundamentals first.

11

00:00:47,040 --> 00:00:52,290

I definitely remember that feeling like trying to memorise a cookbook with a thousand slightly different recipes.

12

00:00:52,620 --> 00:00:59,280

The sheer volume of stuff can push actual problem solving aside instead of a library of facts to memorise.

13

00:00:59,610 --> 00:01:04,140

What if organic chemistry could be more like an operating system? Something logical?

14

00:01:04,350 --> 00:01:11,820

That's the idea behind the source material we're looking at today. It proposes this really, um, elegant alternative, a simple set of priority rules.

15

00:01:12,090 --> 00:01:13,290

Okay. Priority rules.

16

00:01:13,770 --> 00:01:21,340

The goal here, as I understand it, is to learn how to think mechanistically before you even get bogged down in specific named reactions.

17

00:01:21,360 --> 00:01:30,929

Exactly. It's about empowering you, the listener, to look at molecules and predict how they might react based on first principles, even complex stuff.

18

00:01:30,930 --> 00:01:38,480

Maybe reactions you haven't explicitly been taught. That sounds incredibly useful, because organic chemistry isn't just an academic hurdle, is it?

19

00:01:38,490 --> 00:01:46,170

It's. Yeah, well it's everywhere. It really is. It's the basis for drug discovery, materials science, molecular machines, you name it.

20

00:01:46,560 --> 00:01:50,430

But if we're just testing recall, like a lot of current assessments seem to do,

21

00:01:50,640 --> 00:01:54,300

we're not building that deeper critical thinking needed for real innovation.

22

00:01:54,540 --> 00:01:58,020

So instead of learning reaction recipes, mix A and B, get C.

23

00:01:58,170 --> 00:02:04,020

We need a more, um, algorithmic way a system for figuring out those curly arrows, the electron movements.

24

00:02:04,020 --> 00:02:07,380

Yes, these priority rules offer that framework.

25

00:02:07,410 --> 00:02:12,690

They give you a structured way to find the reactive spots and molecules and predict where the electrons are likely to go.

26

00:02:12,960 --> 00:02:17,430

It gives you a starting point, a way into the problem. What's the foundation for these rules?

27

00:02:17,430 --> 00:02:24,550

What do we need to know first? Okay. So to really get these rules we need basically three quick concepts under our belt.

28

00:02:24,570 --> 00:02:31,110

First up inductive effects. We sometimes write that as plus or minus I (-I) okay.

29

00:02:31,230 --> 00:02:34,260

Think of this like a constant tug of war over electrons.

30

00:02:34,500 --> 00:02:42,240

It's based purely on electronegativity. How strongly an atom pulls electrons and it works through the single bonds, the sigma framework.

31

00:02:42,450 --> 00:02:46,890

The more electronegative atom just continuously pulls electron density its way.

32

00:02:46,920 --> 00:02:53,520

Got it like a constant background pool. What's next? Next are mesomeric effects, or sometimes called resonance.

33

00:02:53,820 --> 00:03:00,900

Now, if inductive effects are that quiet tug of war, mesomeric effects are more like a big flowing river of electrons.

34

00:03:00,990 --> 00:03:06,780

Uh, yeah. These involve moving electron pairs through multiple bonds or systems with lone pairs.

35

00:03:06,780 --> 00:03:11,040

The pi system and these are generally more powerful. Oh, much more significant usually.

36

00:03:11,220 --> 00:03:15,270

And they can operate over longer distances within the molecule compared to inductive effects.

37

00:03:15,270 --> 00:03:18,899

Very important okay. Inductive tug of war. Mesomeric River.

38

00:03:18,900 --> 00:03:23,790

What's the third piece? The third piece tackles a common, um, conceptual hang up.

39

00:03:24,210 --> 00:03:28,770

It's about intermediates in a reaction. You often hear textbooks talk about stable intermediates.

40

00:03:28,770 --> 00:03:32,420

Yeah, and that always felt weird to me. If it's stable, why does it keep reacting?

41

00:03:32,430 --> 00:03:35,790

Shouldn't the reaction just stop there? That's the confusion.

42

00:03:36,090 --> 00:03:40,260

So we actually prefer the phrasing energetically accessible intermediates.

43

00:03:40,620 --> 00:03:44,160

The point of a mechanism isn't to make something so stable it stops reacting.

44

00:03:44,580 --> 00:03:49,380

The point is to find the pathway that requires the least energy overall to get from start to finish.

45

00:03:49,650 --> 00:03:52,800

Accessibility, not rock solid stability, is the key.

46

00:03:53,070 --> 00:03:56,700

Energetically accessible. I like that it makes more sense for something that's part of a process.

47

00:03:56,970 --> 00:04:00,120

Nice. Okay. Foundation laid. Let's get into the rules themselves.

48

00:04:01,140 --> 00:04:03,750

It's actually pretty simple when you boil it down.

49

00:04:03,750 --> 00:04:12,480

Deceptively simple really at its heart, any organic reaction mechanism, no matter how complicated it looks, follows three basic conditions.

50

00:04:13,110 --> 00:04:16,800

You need electrons starting somewhere electron rich.

51

00:04:16,950 --> 00:04:23,670

Okay, like a source moving to somewhere electrons pool - their destination. And ending up somewhere, well, somewhere good or stable.

52

00:04:24,060 --> 00:04:29,400

We're going to formalise this with five specific priority rules that you apply in order.

53

00:04:29,610 --> 00:04:35,670

All right let's get into it. Level one priority rules. Rule one where did the electron start finding that electron rich site.

54

00:04:35,790 --> 00:04:40,260

This is the absolute highest priority starting point for drawing that first curly arrow.

55

00:04:40,440 --> 00:04:44,570

And there's a fixed hierarchy okay. Number one. Number one is a negative charge.

56

00:04:44,580 --> 00:04:48,540

Can't get much more electron rich than having an extra electron pair sitting there.

57

00:04:48,540 --> 00:04:54,240

It's basically shouting use me! Make sense? It's next. Number two is an uncharged lone pair.

58

00:04:54,270 --> 00:04:58,170

Think of the electrons on a nitrogen, an ammonia or an oxygen and water.

59

00:04:58,900 --> 00:05:03,670

Very available, but maybe not quite as, uh, pushy as a full negative charge.

60

00:05:03,760 --> 00:05:07,000

Okay. Negative charge then. Lone pair. What follows those?

61

00:05:07,480 --> 00:05:13,680

We get into bonds now, right? Right. Number three is a multiple bond, which we call a pi system.

62

00:05:13,720 --> 00:05:15,400

Think double bonds. Triple bonds.

63

00:05:15,940 --> 00:05:24,250

Those pi electrons are relatively high energy, more exposed, easier to get involved in reactions compared to single bonds.

64

00:05:24,490 --> 00:05:32,200

And the last resort. Number four. The lowest priority starting point is a weak single bond, a sigma bond.

65

00:05:32,740 --> 00:05:39,250

Usually this involves bonds to large, easily polarised atoms like iodine or maybe strained ring systems.

66

00:05:39,580 --> 00:05:45,220

You typically only break these strong sigma bonds if there are no better options available, so it's a strict order.

67

00:05:45,460 --> 00:05:52,000

If you have both a negative charge and a lone pair on the same molecule, the negative charge always dictates where the first arrow starts.

68

00:05:52,000 --> 00:05:57,940

Always. Rule one is absolute and its sequence - negative, charge first, then lone pair, then pi bond and sigma bond.

69

00:05:57,970 --> 00:06:02,110

Got it. So rule one gives us the starting block. Now rule two where do those electrons go?

70

00:06:02,170 --> 00:06:06,430

The electron poor site. The target. Yeah. Where the arrow needs to point. And again there's a priority order.

71

00:06:06,430 --> 00:06:09,610

Highest priority target I'm guessing a positive charge. Exactly.

72

00:06:09,640 --> 00:06:15,050

Number one is a positively charged carbon, uh, carbenium ion or just a proton H^+ .

73

00:06:15,080 --> 00:06:18,730

So these species are seriously electron deficient.

74

00:06:19,030 --> 00:06:22,780

A carbenium ion only has six electrons around it, H^+ has none.

75

00:06:23,050 --> 00:06:29,230

They desperately need electrons to get a full valence shell. Okay, that seems straightforward, but the second priority target.

76

00:06:29,680 --> 00:06:35,889

This is where you mentioned it gets interesting and potentially confusing. It's an atom attached to a positively charged heteroatom.

77

00:06:35,890 --> 00:06:39,760

Correct. Like a hydrogen attached to a positively charged oxygen or nitrogen.

78

00:06:39,760 --> 00:06:43,090

Okay. Hold on. Why not attack the positive heteroatom directly?

79

00:06:43,450 --> 00:06:47,170

Isn't positive the most attractive target? According to priority one here.

80

00:06:47,500 --> 00:06:52,209

This feels like a contradiction. Huh? That's a fantastic question because it hits on a fundamental principle.

81

00:06:52,210 --> 00:07:00,340

Yeah, the octet rule. Remember, atoms like oxygen or nitrogen want eight valence electrons even when they have a positive formal charge,

82

00:07:00,340 --> 00:07:03,850

like an H_3O^+ ion - that oxygen already has a full octet.

83

00:07:03,850 --> 00:07:11,589

Oh, right. Okay. So if your electron pair your nucleophile trying to attack that oxygen directly, you'd be trying to force ten electrons on to it.

84

00:07:11,590 --> 00:07:16,210

That's a huge no-no in chemistry violates the octet rule big time.

85

00:07:16,300 --> 00:07:22,240

So the attack has to happen nearby to sort of relieve the situation without breaking the octet rule.

86

00:07:22,240 --> 00:07:27,160

Precisely. The arrow attacks an adjacent atom, usually a hydrogen or sometimes a carbon.

87

00:07:27,670 --> 00:07:34,510

This allows the electrons in the bond connecting that atom to the positive heteroatom to shift back on to the heteroatom.

88

00:07:34,510 --> 00:07:41,350

Ahhhh, so it neutralises the positive charge by giving the heteroatom its lone pair back, satisfying the octet.

89

00:07:41,470 --> 00:07:44,590

Clever. It's a crucial distinction for getting mechanisms right.

90

00:07:44,770 --> 00:07:51,760

So after positively charged C or H and atoms next to positive heteroatoms, what's the next level of electron poor sight.

91

00:07:52,210 --> 00:07:56,350

That would be number three a partially positive carbon or hydrogen.

92

00:07:56,740 --> 00:07:58,210

The delta plus ones. Right.

93

00:07:58,420 --> 00:08:05,590

These occur when carbon or hydrogen is bonded to something much more electronegative, like oxygen, nitrogen or a halogen like chlorine.

94

00:08:05,590 --> 00:08:11,380

That atom pulls electron density away, leaving the C or H slightly electron poor or delta positive.

95

00:08:11,500 --> 00:08:17,139

Okay, so we have a start - rule 1. A target - rule 2. Now rule 3 where does the electron density actually end up?

96

00:08:17,140 --> 00:08:21,640

The somewhere good rule you called it. This stops us breaking that octet rule exactly.

97

00:08:22,420 --> 00:08:29,139

When the new bond forms, something often has to give way. There are three main outcomes for where that electron density goes.

98

00:08:29,140 --> 00:08:32,980

First one the simplest is if the target was a C plus or H plus.

99

00:08:33,160 --> 00:08:38,980

Rule 2.1, the new bond just forms quenching the charge - mechanism might stop there.

100

00:08:39,370 --> 00:08:45,429

Stable bond formed - done. But if we attacked a delta plus atom like in a carbonyl group, right?

101

00:08:45,430 --> 00:08:50,080

If you attack a delta plus carbon, that's part of a double bond to a heteroatom like C=O,

102

00:08:50,350 --> 00:08:56,379

the electron density often shifts onto that hetero atom by breaking the pi bond, so the C=O becomes C-O minus.

103

00:08:56,380 --> 00:09:00,460

That negative charge on oxygen is pretty stable. That's outcome number two. A third way.

104

00:09:00,490 --> 00:09:04,660

The third way is breaking a single bond, a sigma bond, usually to a heteroatom.

105

00:09:04,660 --> 00:09:05,950

This is what we call a leaving group.

106

00:09:06,250 --> 00:09:11,890

The incoming electrons force the electrons in that sigma bond to leave entirely with the departing atom or group.

107

00:09:12,070 --> 00:09:15,400

Think of Cl leaving. Okay, so the electrons push out another group.

108

00:09:15,640 --> 00:09:19,420

Make sense? That covers the main flow. What about the supporting rules? Rule four and five.

109

00:09:19,720 --> 00:09:23,860

Rule four is charge balance. It's basically just careful accounting.

110

00:09:24,370 --> 00:09:30,460

If a neutral thing like an amine nitrogen using its lone pair forms a bond, it must gain a positive charge.

111

00:09:30,610 --> 00:09:35,590

And conversely, if a positive atom gets its lone pair back, it loses the positive charge.

112

00:09:36,010 --> 00:09:39,639

Forgetting these charges is probably the single most common mistake students

113

00:09:39,640 --> 00:09:45,370

make when drawing mechanisms got to balance the books and rule five iteration.

114

00:09:45,370 --> 00:09:49,450

Yeah, this just acknowledges that reactions often happen in multiple steps.

115

00:09:50,020 --> 00:09:57,669

If your first mechanistic step following rules 1 to 4 creates a new reactive site, maybe that O minus we mentioned or a new positive charge.

116

00:09:57,670 --> 00:10:00,950

Well, that new site might be the starting point for the next step.

117

00:10:00,980 --> 00:10:04,670

So you just loop back, apply rule one again to the new situation. Exactly.

118

00:10:04,670 --> 00:10:07,790

You keep applying the rules sequentially. Find the best electron source.

119

00:10:07,790 --> 00:10:13,640

Find the best target, see where the electrons end up. Balance charge until you reach a stable product or run out of reactive species.

120

00:10:14,090 --> 00:10:17,510

The mechanism unfolds step by step, following these priorities.

121

00:10:17,690 --> 00:10:24,800

Okay, theory is great, but let's see it work. You mentioned an example a key step in making salbutamol the asthma drug.

122

00:10:24,830 --> 00:10:31,220

Yes, a key step in its synthesis is perfect. We start with a molecule that has a primary amine group - an NH_2 .

123

00:10:31,520 --> 00:10:37,909

And we react it with another piece that has a carbon attached to a chlorine atom, sort of like an alkyl chloride derivative.

124

00:10:37,910 --> 00:10:43,370

Right. Step one PR1. The most electron rich spot is the lone pair on the nitrogen of the amine.

125

00:10:43,820 --> 00:10:47,600

That's PR 1 priority 2 - uncharged lone pair.

126

00:10:47,660 --> 00:10:51,560

Okay, where does it go? PR 2. It attacks the carbon attached to the chlorine.

127

00:10:51,890 --> 00:10:56,660

That carbon is electron poor. It's delta plus because chlorine is electronegative.

128

00:10:56,660 --> 00:11:00,350

So that's PR 2, Priority 3 attacking a delta plus carbon.

129

00:11:00,590 --> 00:11:04,610

Ahhh, and because we attacked a delta plus we need PR 3 immediately.

130

00:11:04,610 --> 00:11:13,459

Exactly. To avoid breaking the octet on carbon, the C-Cl bond has to break and those electrons move on to the chlorine atom creating a chloride ion.

131

00:11:13,460 --> 00:11:18,650

And what's left with the main molecule? Well, the nitrogen formed a new bond using its lone pair.

132

00:11:18,650 --> 00:11:23,090

So PR 4 tells us the nitrogen now has a positive formal charge N plus.

133

00:11:23,330 --> 00:11:29,510

And we also have the Cl⁻ floating around. So we have a charged intermediate B⁺ and Cl⁻ that triggers PR 5.

134

00:11:29,750 --> 00:11:36,049

Iterate. Go back to rule one. What's the most electron rich thing now the chloride ions (Cl⁻) it has a negative charge.

135

00:11:36,050 --> 00:11:39,740

So that's PR 1, top priority. Perfect. Now PR 2 where does it go.

136

00:11:40,070 --> 00:11:46,309

It's attracted to the positive nitrogen. But remember it can't attack the N⁺ directly octet rule violation.

137

00:11:46,310 --> 00:11:53,870

It has to attack an adjacent atom. Precisely. In this case, the easiest adjacent atom to attack is one of the hydrogens attached to the N⁺.

138

00:11:54,050 --> 00:11:57,709

The Cl⁻ acts as a base here, so the Cl⁻ plucks off a proton.

139

00:11:57,710 --> 00:12:03,050

Yep, it attacks the H. The electrons that were in the NH bond then move back on to the nitrogen atom.

140

00:12:03,110 --> 00:12:10,459

Uh, which neutralises the positive charge on the nitrogen, and we're left with the final stable organic product plus HCl

141

00:12:10,460 --> 00:12:18,080

essentially. Exactly a clean two step mechanism derived entirely by following the priority rules, including the iteration.

142

00:12:18,260 --> 00:12:22,610

That example also neatly shows the choice that sometimes happens in PR 2, right?

143

00:12:22,880 --> 00:12:27,830

The chloride acted as a base attacking hydrogen, not as a nucleophile attacking carbon.

144

00:12:27,830 --> 00:12:34,820

Correct. The electron rich species often has that choice attack a carbon, be a nucleophile, or attack a proton, be a base.

145

00:12:35,450 --> 00:12:39,470

Now, for beginners, we often just tell them which role our reagent plays to simplify things.

146

00:12:39,500 --> 00:12:41,810

Reduces the cognitive load, as you said, right?

147

00:12:41,960 --> 00:12:49,010

But understanding why something chooses to be a base or a nucleophile, factors like sterics, electronics.

148

00:12:49,310 --> 00:12:53,510

That's where we need a bit more sophistication. That leads us towards level two rules.

149

00:12:53,510 --> 00:12:55,820

Before we get to level two though, let's just connect this back.

150

00:12:56,120 --> 00:13:00,920

These rules feel intuitive, but what's the deeper chemical principle making them work?

151

00:13:01,070 --> 00:13:08,510

Is it related to those orbitals you mentioned earlier? It absolutely is a think of these priority rule as a very useful, um, practical shortcut.

152

00:13:08,750 --> 00:13:15,889

What they're really doing in a rough way is estimating the best possible overlap between the highest occupied molecular orbital,

153

00:13:15,890 --> 00:13:21,800

the HOMO, where the electron pair starts, and the lowest unoccupied molecular orbital, the LUMO or the electron pair is going.

154

00:13:21,860 --> 00:13:28,820

Exactly. The reaction happens most easily when the highest energy electrons can flow into the lowest energy empty spot.

155

00:13:29,330 --> 00:13:35,200

So when we draw an arrow from, say, a lone pair on an oxygen which is in a high energy non-bonding orbital,

156

00:13:35,210 --> 00:13:44,210

the HOMO to an empty p-orbital on a carbocation, the homo, that arrow represents the formation of a new lower energy sigma bonding orbital.

157

00:13:44,210 --> 00:13:49,400

So the rules are basically guiding us to the most energetically favourable HOMO-LUMO interaction,

158

00:13:49,640 --> 00:13:53,030

without needing to calculate the orbitals themselves. That's a great way to put it.

159

00:13:53,030 --> 00:13:59,180

You don't need quantum mechanics to follow the rules, but it's reassuring to know they're grounded in sound physical principles.

160

00:13:59,420 --> 00:14:08,870

Okay, good. Now complexity. What happens in bigger molecules where maybe you have, say, two different lone pairs or two different delta plus carbons.

161

00:14:09,020 --> 00:14:13,970

How do you choose? Uh, yes. Now we enter the realm of selectivity.

162

00:14:14,300 --> 00:14:18,200

This is where you might have multiple sites that fall under the same priority level.

163

00:14:18,200 --> 00:14:23,840

In PR 1 or PR 2, we need level two mechanistic selectivity rules to make the right choice.

164

00:14:23,900 --> 00:14:28,220

Makes sense. Let's start with selectivity. Rule one where the electrons start.

165

00:14:28,400 --> 00:14:34,670

If you have two potential starting points, how do you pick? Okay, here's a slightly counterintuitive one related to electronegativity.

166

00:14:35,000 --> 00:14:42,830

If you're comparing two different atoms, as electron donors say, comparing iodide I versus fluoride F acting as nucleophile, which is better?

167

00:14:42,890 --> 00:14:47,330

Hmm fluorine is more electronegative. So maybe it holds electrons better.

168

00:14:47,420 --> 00:14:54,500

So iodide is better at giving them away. Exactly. It seems backward at first, but the curly arrow means the electrons are leaving.

169

00:14:54,800 --> 00:14:58,430

The less electronegative atom holds its electrons less tightly.

170

00:14:58,530 --> 00:15:02,500

Making it a better donor. So I is actually a much better nucleophile than F.

171

00:15:02,580 --> 00:15:05,639

Interesting. What else affects the starting point choice?

172

00:15:05,640 --> 00:15:14,219

Mesomeric effects again. If a lone pair is heavily involved in resonance or mesomeric stabilisation, it's delocalised.

173

00:15:14,220 --> 00:15:18,570

It's spread out and more stable, making it less available to react. Precisely!

174

00:15:18,570 --> 00:15:22,649

That's why an amide nitrogen is a much poorer nucleophile than an amine nitrogen.

175

00:15:22,650 --> 00:15:27,360

In the amide, that lone pair is busy being shared with the carbon oxygen.

176

00:15:27,360 --> 00:15:32,770

Through the +M effect. It's less available to attack something else. And sterics? bulkiness?

177

00:15:32,820 --> 00:15:36,540

Huge factor, especially for the nucleophile versus base decision.

178

00:15:36,870 --> 00:15:44,489

Generally, less bulky reagents make better nucleophiles because they can physically reach the carbon atom they need to attack, whereas a big, bulky,

179

00:15:44,490 --> 00:15:49,680

electron rich thing might find it really hard to get close to an equally bulky carbon centre

180

00:15:50,070 --> 00:15:56,219

but a small proton, H^+ , on the periphery of a molecule that's much easier to reach.

181

00:15:56,220 --> 00:16:03,660

So very bulky bases like potassium tert-butoxide are great at removing protons, but terrible as nucleophiles.

182

00:16:04,020 --> 00:16:13,800

Sterics favours basicity. OK, so electronegativity, better donor Delocalisation reduces reactivity. And steric bulk favours basicity.

183

00:16:14,490 --> 00:16:18,930

Now for the destination selectivity rule to where the electrons move to.

184

00:16:19,110 --> 00:16:21,300

What if you have multiple electron pore sites?

185

00:16:21,480 --> 00:16:27,360

Here it's mostly about figuring out which site is most electron pore which has the largest delta plus charge,

186

00:16:27,360 --> 00:16:30,959

and that depends on the surrounding atoms entirely. Think about carbonyl groups again ($C=O$).

187

00:16:30,960 --> 00:16:36,450

Is the carbon in a ketone more or less delta positive than the carbon in an aldehyde?

188

00:16:36,750 --> 00:16:43,260

It depends on what else is attached. Alkyl groups are slightly electron donating (+I effect), so they reduce the delta positive character.

189

00:16:43,650 --> 00:16:49,350

Aldehydes only have one alkyl group and one H, while ketones have two, so the aldehyde carbon is slightly more delta positive,

190

00:16:49,350 --> 00:16:52,620

making it generally more reactive towards nucleophile than a ketone carbon.

191

00:16:52,770 --> 00:16:58,799

Correct. It's all about the subtle push and pull of electrons governed by those inductive and mesomeric effects.

192

00:16:58,800 --> 00:17:02,310

And you mentioned a really cool connection here to biology. Yes.

193

00:17:02,310 --> 00:17:10,770

It's fascinating. Why are proteins the workhorses of life built from amide linkages joining amino acids, not ester linkages?

194

00:17:10,770 --> 00:17:15,570

Okay, why? Based on what we just said, the amide nitrogen has +M effect.

195

00:17:15,610 --> 00:17:23,370

Exactly. That nitrogen pushes electron density onto the amide carbonyl carbon, making it significantly less delta positive,

196

00:17:23,490 --> 00:17:32,219

less electron poor than the carbon in an ester group, which only has an oxygen effect, and weaker +M effect pulling density away.

197

00:17:32,220 --> 00:17:39,590

So the amide bond is just... less reactive. Much less reactive towards hydrolysis breakdown by water,

198

00:17:39,590 --> 00:17:45,350

and that stability is essential for proteins to function and persist in the aqueous environment inside cells.

199

00:17:45,740 --> 00:17:51,150

Life literally relies on that subtle difference in reactivity dictated by these electronic effects.

200

00:17:51,170 --> 00:17:57,170

That's incredible. So one more selectivity rule. Rule three - where the electrons end up.

201

00:17:57,530 --> 00:18:00,500

This seemed related to intermediate stability. Yes.

202

00:18:00,650 --> 00:18:06,470

It's about ensuring the reaction proceeds through the most stable or energetically accessible intermediate.

203

00:18:06,740 --> 00:18:11,290

The source material prefers that term over the more technical Hammond postulate.

204

00:18:11,300 --> 00:18:14,300

Why avoid Hammond? It sometimes confuses students.

205

00:18:14,300 --> 00:18:19,190

They hear stable, intermediate and think it means the intermediate is unreactive, which isn't the point.

206

00:18:19,790 --> 00:18:26,600

Energetically accessible focuses on the fact that the pathway to forming that intermediate is lower in energy, meaning the reaction is faster.

207

00:18:26,870 --> 00:18:28,760

So the reaction takes the easiest route,

208

00:18:29,120 --> 00:18:35,270

the one with the most stable checkpoint along the way. Pretty much! Think about adding something like HBr to an outcome.

209

00:18:35,540 --> 00:18:39,530

The first step is the alkene's pi bond attacking the H delta+.

210

00:18:40,010 --> 00:18:44,059

This forms a carbocation intermediate before the Br- attacks,

211

00:18:44,060 --> 00:18:50,240

and there usually two possible carbocations you could form right primary versus secondary or secondary versus tertiary.

212

00:18:50,270 --> 00:18:54,679

Exactly. And the rules predict you'll form the more stable carbocation.

213

00:18:54,680 --> 00:18:58,970

Intermediate tertiary is more stable than a secondary which is more stable than primary.

214

00:18:59,090 --> 00:19:03,410

Why? Because that intermediate is lower and energy easier to form.

215

00:19:03,560 --> 00:19:11,600

That dictates which product you ultimately get. So putting it all together, these level one and level two rules give you a powerful toolkit.

216

00:19:12,050 --> 00:19:18,590

It's less about memorising hundreds of reactions and much more about understanding the fundamental principles of electron movement,

217

00:19:18,590 --> 00:19:23,030

charge, stability, and shape. It lets you work out how molecules are likely to behave.

218

00:19:23,030 --> 00:19:26,989

Just by looking at them. You develop a chemical intuition, a feel for it.

219

00:19:26,990 --> 00:19:30,440

It's like learning the grammar of chemistry, not just vocabulary lists.

220

00:19:30,770 --> 00:19:37,429

That's a great analogy. If curly arrows are the language, these rules are like the phonics or the basic grammar.

221

00:19:37,430 --> 00:19:43,180

They give you the building blocks to become fluent, to read and write chemical reactions with confidence.

222

00:19:43,190 --> 00:19:48,950

So the big takeaway for you listening is that you don't have to feel overwhelmed by organic chemistry.

223

00:19:49,190 --> 00:19:52,040

There's a logical system underneath it all. Absolutely.

224

00:19:52,310 --> 00:19:57,740

These rules provide a step by step method to tackle mechanisms, even ones you've never encountered before.

225

00:19:58,040 --> 00:20:02,120

Okay, final provocative thought time. We've built up this logical system.

226

00:20:02,570 --> 00:20:09,950

The level one rules, for instance, generally predict a mechanism called SN2 attack, and leaving group departure happening at the same time.

227

00:20:09,950 --> 00:20:13,819

Right, that's the standard outcome for many simple cases based on the priority rules.

228

00:20:13,820 --> 00:20:15,500

But chemistry loves exceptions.

229

00:20:15,740 --> 00:20:22,580

There's another common mechanism, SN1, where the leaving group leaves first form a carbocation and then the nucleophile attacks.

230

00:20:22,820 --> 00:20:27,530

The steps are in a different order. Correct. And this is where the rules become even more powerful.

231

00:20:27,590 --> 00:20:30,409

By understanding the standard prediction from the rules,

232

00:20:30,410 --> 00:20:35,690

you can start to analyse why certain molecules or conditions lead to a different outcome like SN1.

233

00:20:35,840 --> 00:20:39,889

So the rules help you understand the exceptions to exactly. They give you the baseline.

234

00:20:39,890 --> 00:20:44,209

When reality deviates, you can ask why did the leaving group leave first here?

235

00:20:44,210 --> 00:20:49,880

What made that carbocation intermediate so accessible even though rule one didn't immediately point that way?

236

00:20:50,570 --> 00:20:56,899

So here's something for you to ponder. Why might the order of steps change if your electron source, your nucleophile,

237

00:20:56,900 --> 00:21:00,740

isn't very strong and the carbocation that would form is particularly stable?

238

00:21:00,890 --> 00:21:05,330

Go look it up. It all loops back to those core principles of stability and reactivity.